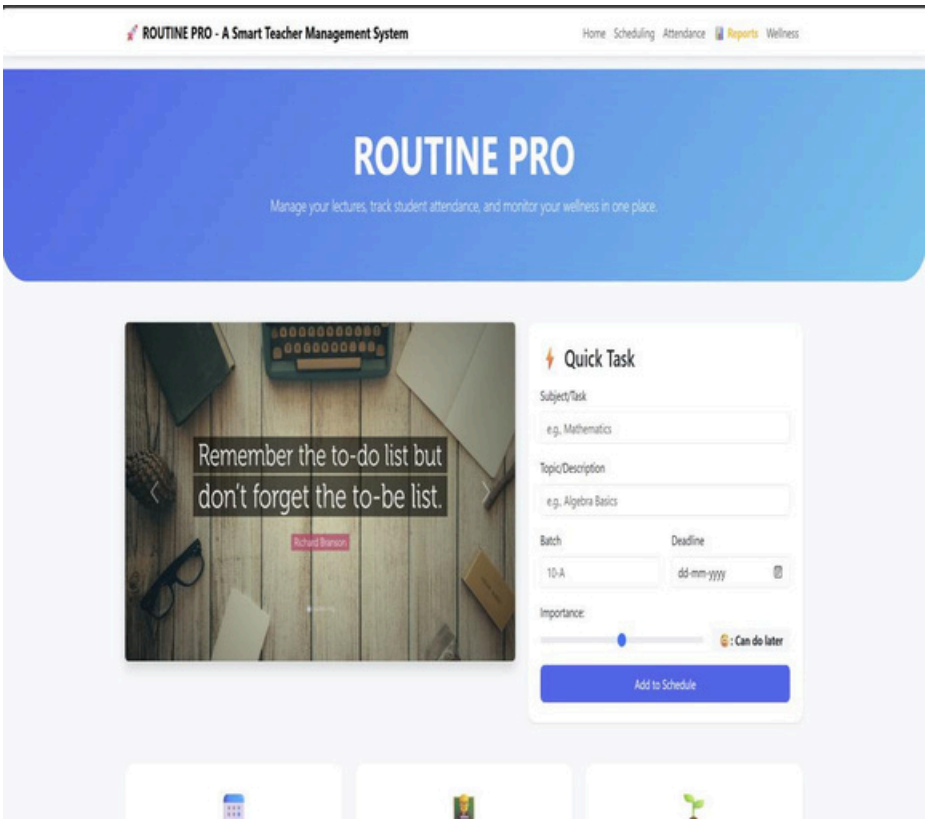

Routine Pro – A Smart Teacher Management System

[Document subtitle]

PROJECT REPORT

Routine Pro is a web-based application designed to help teachers manage lecture schedules, track student attendance, and maintain daily wellness logs in one unified platform.

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Introduction

In modern education systems, teachers are required to manage multiple responsibilities including lecture planning, student attendance tracking, and maintaining personal teaching records. These tasks are often handled using separate tools or manual methods, leading to inefficiency and lack of organization.

This project, Routine Pro, is a web-based application designed to integrate these functionalities into a single platform. It provides a simple and effective solution for teachers to manage their daily academic workflow.

Problem Statement

Teachers face difficulties in:

- Organizing lecture schedules efficiently
- Maintaining accurate student attendance records
- Tracking attendance performance over time
- Keeping personal logs related to teaching and wellness

Existing solutions are either fragmented or overly complex. There is a need for a unified, user-friendly system.

Objectives

The main objectives of this project are:

- To develop a centralized platform for lecture scheduling
- To implement an efficient attendance tracking system
- To provide analytical insights into student attendance
- To include a wellness tracking module for teachers
- To ensure ease of use through a clean and responsive interface

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Methodology

The project follows a modular approach:

- Frontend: Developed using HTML, CSS, and Bootstrap for responsive design

- Backend: Implemented using Python Flask framework
- Database: SQLite database managed using SQLAlchemy ORM

Steps:

1. Requirement analysis of teacher needs
2. Designing database schema
3. Implementing backend logic using Flask
4. Creating user interface with Bootstrap
5. Integrating modules and testing functionality

System Design

Database Models

The system consists of four main entities:

- LectureTask: Stores lecture schedules
- Student: Stores student details
- Attendance: Tracks attendance records linked to students
- Wellness: Stores teacher mood and stress logs

Relationships:

- One student can have multiple attendance records

Features of the System

Lecture Scheduling

- Add subject, topic, batch, deadline, and priority
- View all scheduled lectures
- Delete completed tasks

Attendance Management

- Add students batch-wise
- Mark attendance using checkboxes

- Store attendance by date

Attendance Analytics

- Calculate attendance percentage
- Display results with visual progress bars

Wellness Tracking

- Record mood and stress level
- Maintain a daily journal
- View past wellness logs

Implementation Details

- Flask routes handle different functionalities such as scheduling, attendance, and wellness
- SQLAlchemy is used to manage database operations
- Bootstrap components are used for UI styling
- Forms are used for user input and data submission

Challenges Faced

- Designing relationships between database tables
- Managing attendance updates efficiently
- Handling date-based records
- Ensuring clean and responsive UI
- Debugging form submissions and routing

Results and Output

The system successfully:

- Stores and manages lecture schedules
- Tracks student attendance accurately
- Generates attendance reports with percentages

- Maintains wellness logs for teachers

The application provides a unified dashboard for all functionalities.

Future Improvements

The project can be further enhanced by:

- Adding a secure login and authentication system
- Implementing role-based access (teacher/admin)
- Introducing graphical analytics (charts for attendance trends)
- Enabling data export (CSV/PDF reports)
- Integrating AI-based insights (e.g., identifying low-performing students)
- Deploying the application online for real-world usage

Conclusion

Routine Pro successfully addresses the problem of fragmented teacher management tools by providing a unified and efficient system. The project demonstrates practical application of web development concepts including backend logic, database management, and user interface design.

It also highlights the importance of integrating multiple functionalities into a single platform for improved productivity.

Learning Outcomes

Through this project, the following concepts were learned:

- Flask web development
- Database design using SQLAlchemy
- Frontend development with HTML/CSS/Bootstrap
- CRUD operations and routing
- Problem-solving and debugging
- Real-world application development